Lecture 22: Diffusion model extensions

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain how **latent diffusion** compresses computation via a VAE
2. Describe **classifier-free guidance** and its role in conditional generation
3. Compare U-Net vs Transformer backbones (**DiT** architecture)
4. Understand **flow matching** as a simpler alternative to diffusion
5. Trace the evolution from DDPM to **Stable Diffusion 3**

The pixel problem

Why running diffusion in pixel space is expensive

The continuous diffusion framework from Lecture 21 (DDPM) operates directly on pixel space when applied to images. For a 512×512 RGB image, the denoising network processes tensors of size $512 \times 512 \times 3 = 786{,}432$ values at every step. With 1000 steps, this is prohibitively expensive for high-resolution generation.

The numbers

Resolution	Pixels	U-Net memory	Time per image
64 × 64	12,288	~2 GB	~30 seconds
256 × 256	196,608	~8 GB	~5 minutes
512 × 512	786,432	~32 GB	~20 minutes
1024 × 1024	3,145,728	Not feasible	—

The solution: don't run diffusion in pixel space. Run it in a **compressed latent space**.

Latent diffusion models

Rombach et al. (2022): 'High-Resolution Image Synthesis with Latent Diffusion Models'

Latent diffusion separates the problem into two stages:

1. **Compression**: A pretrained VAE (variational autoencoder) encodes images into a compact latent space (typically 8× spatial compression)
2. **Generation**: Diffusion operates entirely in this latent space

Input Image → VAE Encoder → Latent z (64×64×4) → Diffusion Process → Denoised Latent → VAE Decoder → Output Image (512×512×3)

Why this works

The VAE learns to compress images while preserving perceptually important information. Diffusion in latent space is **~50× cheaper** than in pixel space, with negligible quality loss. This single insight enabled Stable Diffusion — the first open-source, consumer-GPU-capable image generator.

The VAE bottleneck

Compressing images for efficient diffusion

The VAE encoder maps an image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ to a latent $\mathbf{z} \in \mathbb{R}^{h \times w \times c}$ where $h = H/f$, $w = W/f$, and f is the downsampling factor (typically $f = 8$).

Stable Diffusion's VAE

Property	Value
Input resolution	512 × 512 × 3
Latent resolution	64 × 64 × 4
Compression ratio	48× (786K → 16K values)
VAE training	Perceptual loss + adversarial loss
Reconstruction quality	Near-lossless for natural images

The VAE is trained once and frozen. The diffusion model only ever sees latents — it never touches pixel space during training or sampling.

Text conditioning with cross-attention

How text controls image generation

To generate images from text prompts, latent diffusion adds **cross-attention** layers to the U-Net. At each spatial resolution:

1. The text prompt is encoded by a text encoder (e.g., CLIP) into a sequence of embeddings
2. The U-Net's intermediate features serve as **queries**
3. The text embeddings serve as **keys** and **values**
4. Cross-attention allows each spatial location in the image to attend to relevant words

Remember...

- **CLIP** (Contrastive Language-Image Pre-training; Radford et al., 2021): A model trained on 400M image-text pairs to learn a **shared embedding space** where images and their captions are nearby. Used throughout diffusion systems as the text encoder that "understands" prompts.

How 'a cat wearing a hat' becomes an image

The word "cat" activates high attention weights in the spatial region where the cat is being generated. The word "hat" activates attention weights near the top of the cat region. This spatial-linguistic binding is learned entirely from image-caption pairs during training.

Classifier-free guidance

Ho & Salimans (2022): controlling generation quality

Classifier-free guidance (CFG) is a technique for improving the alignment between text prompts and generated images. During training, the text condition is randomly dropped (replaced with an empty prompt) some fraction of the time. At inference, the model makes two predictions:

- **Conditional**: $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c)$ — with the text prompt
- **Unconditional**: $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing)$ — without the text prompt

The final prediction extrapolates *away* from the unconditional toward the conditional:

$$\tilde{\boldsymbol{\epsilon}}_\theta = (1 + w)\, \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c) - w\, \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing)$$

where w is the **guidance scale** (typically 7–15).

The guidance scale tradeoff

What different guidance scales produce

Guidance scale w	Effect	Quality vs diversity
0	Pure unconditional (ignores prompt)	Maximum diversity, no text alignment
1	Standard conditional generation	Moderate alignment
7–10	Recommended range	Strong alignment, good diversity
15–20	Over-guided	Very literal, but saturated/artificial
50+	Extreme	Severe artifacts, mode collapse

The intuition

Think of CFG as asking: "What's different about images that match this prompt versus random images?" Then amplifying that difference. Higher guidance = more amplification = more faithful to the prompt but less natural variation.

Diffusion Transformer (DiT)

Peebles & Xie (2023): replacing U-Net with Transformer

The Diffusion Transformer (DiT) replaces the U-Net backbone with a standard Vision Transformer (ViT):

1. **Patchify**: Divide the latent into non-overlapping patches (e.g., 2×2)
2. **Flatten**: Treat patches as a sequence of tokens (just like ViT treats image patches)
3. **Process**: Apply standard Transformer blocks with self-attention
4. **Unpatchify**: Reshape back to spatial dimensions

Why replace U-Net?

Transformers scale better than U-Nets. DiT-XL/2 (675M parameters) achieves a new state-of-the-art FID of 2.27 on ImageNet, beating all previous diffusion models. More importantly, DiT shows **clean scaling behavior** — larger models consistently produce better results, with no architectural bottlenecks.

adaLN-Zero: conditioning in DiT

Adaptive layer normalization for timestep and class conditioning

DiT conditions on timestep and class label using **adaptive Layer Normalization (adaLN-Zero)**:

1. The timestep and class embeddings are combined and projected to produce **scale** (γ) and **shift** (β) parameters
2. These modulate the LayerNorm output: $\text{adaLN}(\mathbf{h}) = \gamma \odot \text{LN}(\mathbf{h}) + \beta$
3. Additionally, a **gating parameter** α scales the residual connection, initialized to zero

Why 'Zero'?

Initializing the gating parameter $\alpha = 0$ means each Transformer block initially acts as an **identity function**. This makes training stable even for very deep models — the network starts by doing nothing and gradually learns to denoise. This is the same principle behind residual learning (He et al., 2016).

Flow matching

Lipman et al. (2023): a simpler framework

Flow matching offers a cleaner mathematical framework than DDPM. Instead of a discrete chain of noising steps, flow matching defines a **continuous path** from noise to data using an ordinary differential equation (ODE):

$$\frac{d\mathbf{x}}{dt} = v_\theta(\mathbf{x}_t, t)$$

where v_θ is a neural network that predicts the **velocity** (direction and speed) of the flow at each point.

Remember...

- **ODE** (Ordinary Differential Equation): An equation describing how a quantity changes over time via a deterministic rule — given the current state, the next state is fully determined
- **SDE** (Stochastic Differential Equation): Like an ODE but with a random noise term — the path from noise to data has some randomness at each step

Flow matching vs DDPM

Key differences

	DDPM	Flow matching
Path type	Stochastic (SDE)	Deterministic (ODE)
Training	Predict noise ϵ	Predict velocity v
Interpolation	$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$	$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\boldsymbol{\epsilon}$
Simplicity	Requires noise schedule design	Schedule-free

The intuition

Flow matching asks: "What's the simplest path from noise to data?" Instead of designing a complex noise schedule and learning to reverse it, we define a straight interpolation and learn the velocity field that moves along it. The math is simpler, the training is more stable, and generation is faster.

Rectified flow

Straight paths from noise to data

Rectified flow (Liu et al., 2023, ICLR) uses the simplest possible interpolation — a straight line between the data point and a noise sample:

$$\mathbf{x}_t = (1 - t)\,\mathbf{x}_0 + t\boldsymbol{\epsilon}$$

The velocity along this path is constant: $v = \boldsymbol{\epsilon} - \mathbf{x}_0$

Why straight paths matter

Straight paths are the shortest paths between noise and data. Since they don't curve, they can be accurately simulated with **fewer ODE solver steps** — as few as 1–4 steps with distillation. This is the foundation of Stable Diffusion 3's fast generation.

Stable Diffusion 3: putting it all together

Esser et al. (2024): MMDiT architecture

Stable Diffusion 3 combines all of the extensions we've discussed:

Component	Choice	Why
Latent space	VAE with 16-channel latent	Higher capacity than SD 1.x (4 channels)
Backbone	**MMDiT** (multimodal DiT)	Transformers scale better than U-Net
Text encoders	CLIP-L + CLIP-G + T5-XXL	Three encoders for rich text understanding
Conditioning	Joint attention (text + image tokens)	Text and image tokens attend to each other
Training framework	**Rectified flow**	Simpler than DDPM, faster sampling
Guidance	CFG with dynamic rescaling	Better prompt adherence

The trend

Architecture evolution summary

From DDPM to SD3 in four years

Year	Model	Key innovation
2020	DDPM	Practical diffusion with simplified loss
2021	Improved DDPM	Cosine schedule, learned variance
2022	Latent Diffusion / SD 1.x	VAE compression, cross-attention conditioning
2022	CFG	Guidance scale for prompt adherence
2023	DiT	Transformer backbone, clean scaling
2023	Flow matching	Simpler ODE framework, straight paths
2024	SD3 / MMDiT	All of the above, unified

The takeaway

The field progresses by **composing** innovations, not replacing them. Each extension addresses a specific

Discussion

Questions to consider

1. **Latent vs pixel space**: Latent diffusion trades exact pixel control for speed. Are there tasks where pixel-space diffusion would be strictly better? What information might the VAE discard?
2. **Guidance as amplification**: CFG amplifies the difference between conditional and unconditional predictions. Is this analogous to anything in human cognition — e.g., how we exaggerate features when imagining something vividly?
3. **U-Net vs Transformer**: DiT replaced U-Net because Transformers scale better. But U-Net's inductive biases (locality, skip connections) seem useful for spatial data. Is there a hybrid that gets the best of both?
4. **Simplification trend**: DDPM → DDIM → Flow matching → Rectified

Take-home messages

Think about it...

- The key bottleneck in high-resolution generation wasn't the diffusion process itself — it was **where** you run it. Compressing to latent space (via a VAE) made consumer-GPU generation possible.
- Classifier-free guidance shows that **controlling** generation is as important as generation itself — and the trick is surprisingly simple: learn what the conditional and unconditional outputs look like, then amplify the difference.
- The field evolves by **composing** innovations (latent space + CFG + DiT + flow matching), not replacing them. Each addresses one specific limitation.

Further reading

Further reading

Rombach et al. (2022, *CVPR*) "High-Resolution Image Synthesis with Latent Diffusion Models" — Latent diffusion and the foundation of Stable Diffusion.

Ho & Salimans (2022, *arXiv*) "Classifier-Free Diffusion Guidance" — The guidance technique used in virtually all modern diffusion systems.

Peebles & Xie (2023, *ICCV*) "Scalable Diffusion Models with Transformers" — DiT: replacing U-Net with Transformers.

Lipman et al. (2023, *ICLR*) "Flow Matching for Generative Modeling" — A simpler, ODE-based alternative to diffusion SDEs.

Esser et al. (2024, *arXiv*) "Scaling Rectified Flow Transformers for High-Resolution Image Synthesis" — Stable Diffusion 3 and the MMDiT architecture.

Questions?

Email

Discord

Office hours

Up next...

Diffusion applications: text-to-image, text-to-video, and the ethics of generative AI